FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Abdominal & Genito-urinary Examination

Clinical skill | Week 2 · Diarrhoea, fluids, HIV & anaemia | Dr Namugga, Dr Aliku

Examining a child's abdomen and genitalia is about winning the child's trust first (warm hands, the mother's lap, watching the face), then a systematic look–feel–percuss–listen with age-specific normals (a soft liver edge up to 2 cm below the costal margin is normal in infants), always completed by the groins, genitalia, growth, blood pressure and urine dipstick, with a chaperone and without ever forcibly retracting a foreskin or doing an internal examination of a prepubertal child.

Must know before the ward round

1. Order: introduce, consent and assent, chaperone, hand hygiene, warm hands, expose from nipples to knees (preserve dignity), general examination first, then inspect, palpate (light then deep, start away from pain, watch the face), percuss, auscultate, then groins, genitalia, anus (inspect), and complete with growth, BP and urine dipstick.
2. A soft liver edge up to 2 cm below the right costal margin is normal in infants and young children (up to about 3 cm in newborns); measure the liver SPAN by percussion to tell true enlargement from a liver pushed down by hyperinflated lungs.
3. A spleen tip may be palpable in healthy neonates and some infants; otherwise a palpable spleen is abnormal. Grade malarial splenomegaly by the Hackett scale (0–5).
4. Spleen vs left kidney: spleen has a notch, moves down and medially toward the RIF with inspiration, you cannot get above it, and it is dull to percussion; a kidney is ballotable, you can get above it, and it is resonant (overlying bowel).
5. Wilms tumour: smooth flank mass in a well child, does not cross the midline. Neuroblastoma: irregular mass crossing the midline in an unwell child. Burkitt lymphoma: rapidly growing abdominal mass ± jaw swelling (Uganda).
6. Big liver and spleen in Uganda: malaria, schistosomiasis (S. mansoni), sickle cell disease, HIV, TB, Burkitt lymphoma/leukaemia, visceral leishmaniasis (Karamoja), heart failure, typhoid.
7. Inguinal hernia: you cannot get above it, cough/cry impulse, usually reducible; does not transilluminate (except infant bowel can). Hydrocele: you CAN get above it, transilluminates, testis may be impalpable within it. All infant inguinal hernias need prompt herniotomy.
8. Undescended testis: examine warm, milk from the groin down; retractile testes can be brought to the scrotum and stay; refer for orchidopexy if not descended by 6 months (operate at 6–12 months, by 18 months at the latest).
9. Acute scrotum = testicular torsion until proven otherwise: high-riding, horizontal testis, absent cremasteric reflex; surgery within 6 hours saves 80–100% of testes. Do not wait for an ultrasound.
10. A non-retractile foreskin is normal (fully retractable in about 10% at 1 year, 50% at 10 years, 99% by 17 years); never forcibly retract. Hypospadias: do NOT circumcise (the foreskin is needed for repair).
11. Female genitalia: inspection only, child supine in frog-leg position or on the mother's lap, with a chaperone. Never a speculum or digital examination in a prepubertal girl; suspected abuse → senior, child protection, PEP within 72 h.
12. Puberty (Tanner): girls start with breast buds at 8–13 years, boys with testicular volume ≥ 4 mL at 9–14 years. Precocious: < 8 girls, < 9 boys. Delayed: no signs by 13 girls, 14 boys.

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1. Purpose and Indications

The paediatric abdominal and genito-urinary (GU) examination is the **adult examination adapted to a frightened child**. You use the same sequence (inspect, palpate, percuss, auscultate), but you must:

- **Gain trust first** (play, warm hands, examine on the mother's lap if needed), because a crying child tenses the abdomen and you will feel nothing.
- **Know age-specific normals**: a protuberant abdomen, a palpable liver edge and a non-retractile foreskin are normal in young children.
- **Always examine the groins and genitalia** (hernias, undescended testes, torsion, hypospadias) and **complete** the examination with **growth, blood pressure and a urine dipstick**.
- **Protect the child**: chaperone, privacy, consent and assent, and no internal examination of a prepubertal child.

In Uganda the findings carry big diagnoses: **hepatosplenomegaly** (malaria, schistosomiasis, sickle cell disease, HIV, Burkitt lymphoma), **abdominal masses** (Burkitt, Wilms, neuroblastoma), **ascites** (nephrotic syndrome, endomyocardial fibrosis, TB) and **the acute scrotum**.

> CLINICAL PEARL
> The one thing never to forget: **watch the child's face, not your hand.** A grimace or a pause in crying tells you where the tenderness is far better than a 3-year-old's words. And if a boy has abdominal pain, **look at the testes**: torsion often presents as "tummy ache".

Indications

- Abdominal pain, vomiting, diarrhoea, constipation, abdominal distension or mass.
- Jaundice, pallor, fever without a focus (spleen, liver, kidney tenderness), failure to thrive, oedema (ascites).
- GU symptoms: dysuria, haematuria, enuresis, poor stream.
- Scrotal or groin swelling or pain, ambiguous genitalia, undescended testes, pubertal concerns.
- Suspected abuse (examination by trained seniors).
- Every **routine full paediatric examination** and newborn check.

2. Contraindications and Precautions

There is no absolute contraindication to examining a child's abdomen, but the examination is **modified or deferred** in these situations:

- **Shock, airway compromise, convulsions**: resuscitate first (ETAT); a full abdominal examination waits. A shocked child with a distended abdomen needs **resuscitation and a surgeon**, not a long examination.
- **Suspected Wilms tumour or neuroblastoma**: **palpate gently and once** (one examiner); repeated vigorous palpation risks tumour rupture and spread.
- **Suspected sexual abuse**: inspect only what is needed for immediate care; the **forensic examination is done once by a trained clinician** with a chaperone and documentation.

- **Distressed child**: pause, play, examine on the mother's lap, or come back after analgesia or when asleep (a sleeping infant is the best abdomen to palpate).
- **Digital rectal examination**: **not routine** in children; only on a senior's indication, with the little finger in infants.
- **Prepubertal girls**: **inspection only**; never a speculum or digital vaginal examination on the ward.
- **Foreskin**: never forcibly retract.
- **Rebound tenderness**: do not elicit it deliberately; use cough, hop or percussion tenderness instead.

Pain relief early: analgesia does **not** mask surgical signs. Give paracetamol **15 mg/kg** orally or IV; morphine **0.1 mg/kg IV**, titrated, for severe pain, by a senior.

3. Relevant Anatomy and Normal Values

Feature	Normal in children	Why it matters
Abdominal shape	Toddlers have a **protuberant abdomen** with lumbar lordosis ("pot belly") when standing; flattens by school age	Do not call it distension unless tense, shiny, or with dilated veins or fluid
Breathing	Infants are **abdominal (diaphragmatic) breathers**	A still abdomen in a breathing infant suggests **peritonitis**
Liver	Relatively large; **edge palpable 1–2 cm** below the right costal margin in infants and toddlers (up to about 3 cm in newborns), soft and sharp	Measure the **span by percussion**; a firm, nodular or > 2 cm liver is abnormal (hyperinflation from bronchiolitis or asthma pushes a normal liver down)
Liver span (percussion, midclavicular line)	About **4.5–5 cm** in the first week of life; about **7–8 cm (boys)** and **6–6.5 cm (girls)** by 12 years	A large span confirms hepatomegaly
Spleen	Tip palpable in some **healthy neonates and young infants**; otherwise not palpable	Palpable spleen beyond infancy = splenomegaly until proven otherwise
Kidneys	**Lower poles may be felt** in neonates; rarely felt later	Enlarged kidney: hydronephrosis, Wilms, polycystic disease
Bladder	**Abdominal organ in infants** (palpable or percussible above the pubis when full)	A persistently palpable bladder in a boy with a poor stream = **posterior urethral valves**
Umbilicus	**Umbilical hernia is common** in African infants; most close spontaneously by 4–5 years	Reassure; incarceration is rare
Recti	**Divarication of recti** is normal in infants	Not a hernia
Testes	Descend by birth in term boys; may complete descent by **3 months**; the **cremasteric reflex** makes them **retractile** from about 6 months to puberty	Examine warm and relaxed; distinguish retractile from undescended
Foreskin	Non-retractile at birth; retractable in about **10% at 1 year, 50% at 10 years, 99% by 17 years**	Never forcibly retract; ballooning on voiding is normal
Penis	Micropenis = stretched length **< 2.5 cm** at term	Ambiguous genitalia work-up
Female genitalia	Newborns may have **swollen labia, white discharge or a small withdrawal bleed** from maternal oestrogen; the prepubertal hymen is thin and sensitive	Reassure; prepubertal girls: inspection only
Abdominal aorta	Pulsation is normal in thin children	Not an aneurysm

Respiratory rate context (lower-lobe pneumonia presenting as abdominal pain): fast breathing is ≥ 60 (< 2 months), ≥ 50 (2–12 months), ≥ 40 (1–5 years).

Hackett grading of splenomegaly (malaria surveys)

Grade	Spleen tip position (child supine, along the line toward the umbilicus)
0	Not palpable, even on deep inspiration
1	Palpable only on deep inspiration
2	Palpable on normal breathing, but not beyond a horizontal line **halfway between the costal margin and the umbilicus**
3	Beyond that line but **not below the umbilicus**
4	**Below the umbilicus** but not beyond a line halfway between the umbilicus and the pubic symphysis
5	Beyond that line (toward the pubis or right iliac fossa)

On the ward, also record the size in **cm below the left costal margin**, measured along the line toward the right iliac fossa (the axis of splenic enlargement).

Tanner (sexual maturity) stages, simplified

Stage	Boys: genitalia	Girls: breasts	Pubic hair (both)
1	Prepubertal; testes < 4 mL	Prepubertal (papilla only)	None
2	**Testes ≥ 4 mL** (first sign), scrotum reddens and thins	**Breast bud** (first sign)	Sparse, long, slightly pigmented at the base of the penis or labia
3	Penis lengthens	Breast and areola enlarge, no separation of contour	Darker, coarser, curled, spreading over the pubis
4	Penis broadens, glans develops; testes 10–15 mL	Areola and papilla form a **secondary mound**	Adult type, not on the thighs
5	Adult; testes 15–25 mL	Adult contour	Adult, spreading to the medial thighs

Puberty starts at **8–13 years in girls** (breast buds; menarche about 2–2.5 years later) and **9–14 years in boys** (testicular enlargement). **Precocious**: before 8 (girls) or 9 (boys). **Delayed**: no signs by 13 (girls) or 14 (boys).

4. Equipment and Preparation

Equipment

Item	Use
Hand hygiene (soap or alcohol rub), **warm hands**	Infection control; a relaxed abdomen
Toys, a light, a pen torch	Distraction; **transillumination** of scrotal swellings
Tape measure (non-stretch)	Abdominal girth (ascites), liver and spleen size in cm, head circumference
Weighing scale, length board or stadiometer, MUAC tape	Growth and nutrition (part of every GI examination)
Paediatric BP cuff (bladder 80–100% of arm circumference, width ≥ 40%)	Hypertension (AGN, Wilms, renal disease, coarctation)

Item	Use
Stethoscope (paediatric diaphragm)	Bowel sounds, bruits
Gloves, lubricant	Genital and perianal inspection; rarely rectal examination (senior)
Urine dipstick and specimen pot	Haematuria, proteinuria, glucose, nitrites and leucocytes
Orchidometer (Prader beads), Tanner charts	Testicular volume and pubertal staging (adolescents)
WHO growth charts, child health card	Plot and interpret
Chaperone (nurse or parent)	Mandatory for genital and breast examination

First 5 minutes: is the child sick?

1. **Quick ETAT look**: airway, breathing, circulation (capillary refill, pulse), consciousness, convulsions, severe dehydration. Emergency signs come first (section 2).
2. **Pain relief early** (section 2).
3. **Privacy and warmth**: screens, a warm room, keep the child covered where possible.

Focused history (what to ask before you examine)

Ask	Why it matters
Main complaint and duration	Acute (surgical?) versus chronic (organomegaly, mass)
Pain: site, onset, character, colicky versus constant, radiation, relation to meals, stools, urine	Colicky + pallor + red-currant jelly = intussusception; RIF + fever = appendicitis; loin + fever = pyelonephritis
Vomiting (bilious? projectile?)	Bilious = obstruction (malrotation and volvulus); projectile non-bilious at 2–8 weeks = pyloric stenosis
Stools: diarrhoea, blood, constipation, worms passed	Dysentery, intussusception, constipation (faecal mass), Ascaris obstruction
Urine: dysuria, frequency, colour (cola = haematuria; tea = conjugated bilirubin; red = haemoglobinuria or blackwater), stream, enuresis, oliguria	UTI, AGN, posterior urethral valves, malaria haemoglobinuria
Swelling: abdomen (gradual versus rapid), face and feet (oedema), groin or scrotum (comes and goes with crying?)	Ascites or nephrotic syndrome, Burkitt (rapid), hernia (intermittent)
Jaundice, pallor, fever, weight loss, night sweats	Liver disease, haemolysis (SCD), malignancy, TB
Bleeding, bruising, bone pain	Leukaemia
Diet, growth, development, feeding	Malnutrition, chronic disease
Past history: SCD, malaria episodes, HIV status, TB contact, previous surgery, neonatal course (umbilical catheter)	Causes of organomegaly; portal vein thrombosis and portal hypertension; adhesions after surgery
Residence and water contact (lake shores, rivers; Karamoja)	**Schistosomiasis**, **visceral leishmaniasis**
Immunisation (UNEPI) and **family history** (SCD, polycystic kidneys)	Hepatitis B, hereditary causes
Genital concerns: undescended testes noticed at birth, swelling, circumcision plans, ambiguous genitalia, discharge, pubertal timing, possible abuse (behaviour change, bleeding, discharge in a young girl)	Timing of surgery; hypospadias before circumcision; safeguarding

Preparation checklist

- **Consent and assent**: verbal consent from the parent and **assent** from the child in words the child understands; explain specifically **before** examining the genitalia.
- **Chaperone** for genital examination (always), and for any examination of an adolescent. Record the chaperone's name.
- **Positioning and exposure** by age are in section 5, step A.

5. Procedure Step by Step

The sequence follows Hutchison's Clinical Methods (24th edition), adapted for children: be **opportunistic** (observe first, palpate while the child is calm, auscultate while quiet, leave distressing parts to last), use play and distraction, and judge each finding against **age-specific normals** (section 3).

A. Introduction, consent and preparation

1. **Introduce yourself** to the parent and child, confirm the child's name and age, and explain what you will do in simple words.
2. **Consent** from the parent, **assent** from the child; **offer a chaperone** (mandatory before the genitalia).
3. **Hand hygiene**; **warm your hands** (rub them, or warm them under the armpit).
4. **Position**:
    - **Infant**: supine on the couch or on the **mother's lap** (lying across her knees), possibly while breastfeeding or with a dummy or toy.
    - **Toddler**: on the mother's lap facing you, or supine with the mother at the head end holding hands; **distract** (bubbles, toy, "count the buttons").
    - **Older child or adolescent**: supine, **one pillow**, arms by the sides, **knees slightly flexed** to relax the abdomen.
5. **Exposure**: from the **nipples to the knees** for a full inspection of abdomen, groins and genitalia, but in practice **expose the abdomen first** and the genitalia **only when you examine them** (dignity; older children keep underwear on until then).
6. **Kneel or sit** so your eyes are at the level of the abdomen.

B. General inspection from the end of the bed

1. **Well or unwell**; comfortable, **lying still** (peritonitis) or **writhing** (colic); alert; interaction.
2. **Colour**: pallor, jaundice, cyanosis.
3. **Hydration**: dry mouth, sunken eyes, skin pinch.
4. **Nutrition**: wasting, oedema; **dysmorphism**.
5. **Vital signs**: temperature, heart rate, respiratory rate, capillary refill, **BP with the correct cuff**.
6. **Growth**: weight, length or height, head circumference (< 2 years), **MUAC**; **plot Z-scores**.

C. Hands

- Clubbing (chronic liver disease, IBD, cystic fibrosis, cyanotic heart disease), koilonychia, leukonychia (hypoalbuminaemia), palmar erythema, **pallor of the palmar creases**.

D. Face, eyes and mouth

1. **Eyes**: **pallor** (conjunctivae), **jaundice** (sclerae, in daylight), periorbital puffiness (nephrotic syndrome, AGN), raccoon eyes (neuroblastoma).
2. **Face**: frontal bossing and maxillary overgrowth (SCD), **jaw swelling** (Burkitt).
3. **Mouth**: thrush (HIV), ulcers, angular stomatitis, dental caries, **parotid enlargement** (HIV).

E. Neck, nodes, skin and oedema

1. **Lymph nodes**: cervical, axillary and inguinal lymphadenopathy; **JVP** in older children.
2. **Skin**: spider naevi, bruising or petechiae (leukaemia, liver disease), scratch marks (cholestasis), rashes (HSP on buttocks and legs).
3. **Oedema**: feet, sacrum, face.

F. Abdomen: inspection (stand back, then at the level of the abdomen)

1. **Shape**: flat, scaphoid (diaphragmatic hernia in a newborn; dehydration), protuberant (normal toddler), **distended** (the 5 Fs: fluid, flatus, faeces, fat, fetus in adolescents; plus **organomegaly and masses**).
2. **Movement with breathing**: moves freely, or held still (peritonitis).
3. **Visible peristalsis**: waves from the **left upper quadrant to the right** in pyloric stenosis (watch after a test feed); generalised in obstruction.
4. **Skin**: scars, striae (steroids, Cushing), **dilated veins** (portal hypertension: flow away from the umbilicus, caput medusae; IVC obstruction: flow upward), bruising (trauma, abuse), cutaneous signs.
5. **Umbilicus**: hernia (size of the fascial defect), granuloma or discharge (newborn), everted (ascites).
6. **Visible masses**, asymmetry (Wilms bulge in the flank), **divarication of recti** (normal in infants).
7. **Groins**: swelling at rest and when the child cries or coughs.

G. Abdomen: palpation

1. **Ask "where does it hurt?"** (point with one finger); start in the **quadrant farthest from the pain**.
2. **Light palpation** of all 9 regions with the flat of the hand, **watching the face**: tenderness, guarding (voluntary versus involuntary), rigidity. In young children, **palpate with the child's own hand under yours**, or palpate with the stethoscope while "listening".
3. **Deep palpation** of all 9 regions: masses, faecal loading (left iliac fossa, indentable).
4. **Peritonism**: avoid deliberate rebound; instead ask the child to **cough, hop or jump** (pain = peritonism), or tap the heel; **percussion tenderness** is gentler.
5. **Liver**: start in the **right iliac fossa** (a large liver starts low), fingers pointing upward or the radial border of the index finger parallel to the costal margin; move up 1–2 cm with each breath until the edge meets your hand. Describe: **cm below the costal margin in the midclavicular line**, edge (sharp or rounded), surface (smooth or nodular), consistency (soft, firm, hard), tenderness, pulsatility (tricuspid regurgitation). In infants, use a **light** touch: the edge is soft and easily missed.
6. **Spleen**: start in the **right iliac fossa**, move diagonally toward the **left hypochondrium** with each breath; if not felt, roll the child **toward you onto the right side** and palpate again with your left hand supporting the left lower ribs. Describe size (cm along its axis, Hackett grade), notch, consistency, tenderness.
7. **Kidneys**: **bimanual (ballotting)**: one hand behind the loin, one in front; push up with the posterior hand to "bounce" the kidney against the anterior hand. Compare sides.
8. **Bladder**: suprapubic mass that is dull and that you cannot get below; recheck after voiding.
9. **Any mass**: **site, size (cm), shape, surface, edge, consistency, tenderness, mobility (with breathing, side to side), can you get above or below it, does it cross the midline, percussion note, pulsatility, bruit, transillumination**.
10. **Abdominal aorta**: pulsation is normal in thin children.

H. Abdomen: percussion

1. **Liver span**: percuss down the right midclavicular line from the chest (resonant) to the upper border of dullness, and up from the abdomen to the lower border; measure the span in cm.
2. **Spleen**: percuss **Traube's space** (left lower anterior chest) and the left flank; dullness supports splenomegaly.
3. **Masses and bladder**: dull (solid organ, full bladder) versus resonant (bowel in front, for example a kidney).

4. **Ascites**: **shifting dullness** (percuss from the midline to the flank until dull, keep a finger there, roll the child away from you, wait 30 seconds, and percuss again: the dull area becomes resonant). **Fluid thrill** only in tense, large ascites (needs an assistant to place the edge of a hand on the midline). Measure **abdominal girth** at the umbilicus daily.

I. Abdomen: auscultation

1. **Bowel sounds**: listen for up to 1–2 minutes beside the umbilicus. **Absent** (ileus, peritonitis), **hyperactive** (gastroenteritis), **high-pitched tinkling** (obstruction).
2. **Bruits**: renal (over the flanks or above the umbilicus) in a hypertensive child; hepatic (rare).
3. **Succussion splash** (gastric outlet obstruction): gently rock the child.

J. Groins and femoral pulses

1. Inspect the **inguinal regions** with the child standing (older child) or crying (infant) for a swelling.
2. **Hernia**: swelling in the groin or scrotum, **cannot get above it**, cough or cry **impulse**, usually reducible (gently, toward the internal ring).
3. **Femoral pulses**: compare with the right brachial or radial (**radio-femoral delay** or weak femorals = **coarctation of the aorta**).
4. **Inguinal lymph nodes**.

K. Male genitalia (chaperone, warm hands; a toddler sitting **cross-legged ("frog" position)** or squatting relaxes the cremaster)

1. **Penis**: size (micropenis: stretched length < 2.5 cm at term), position of the **urethral meatus** (ventral = **hypospadias**, dorsal = epispadias), **chordee** (ventral curvature), **hooded foreskin** (dorsal hood with hypospadias), circumcision status, inflammation (balanitis), **paraphimosis** (retracted, swollen foreskin trapped behind the glans).
2. **Foreskin**: **do not retract**; note ballooning, scarring of the tip (true phimosis = scarred white ring, balanitis xerotica obliterans), discharge.
3. **Scrotum**: size, symmetry, rugae (underdeveloped on the side of an undescended testis), colour (red or blue in torsion), swelling.
4. **Testes**:
   - Before touching the scrotum, place your fingers at the **inguinal canal** and **"milk" downward** toward the scrotum (prevents the testis retracting upward).
   - Palpate each testis between thumb and fingers: **present?**, size (orchidometer in adolescents), consistency, tenderness, **lie** (vertical normal; horizontal and high-riding in torsion), epididymis (posterolateral), a **"blue dot"** at the upper pole (torsion of the appendix testis).
   - **Retractile** testis: can be brought down to the bottom of the scrotum and **stays there** briefly: normal. **Undescended**: cannot be brought into the scrotum (or springs straight back). **Impalpable**: not felt anywhere (abdominal, absent).
   - **Cremasteric reflex** (stroke the inner thigh; the ipsilateral testis rises): **absent in torsion**.
5. **Scrotal swelling**: can you **get above it**? **Transilluminate** with a torch in a darkened room (hydrocele glows; hernia, tumour and haematoma do not, although infant bowel can transmit some light). Is the testis separate from it?

L. Female genitalia (inspection only; chaperone; parent present)

1. Position: **supine "frog-leg"** (soles of the feet together, knees apart) on the couch or the **mother's lap**; or knee–chest for a brief look in an older child.
2. **Gentle labial separation** (fingers on the labia majora, drawing slightly **outward and downward**) to see the introitus.

3. Look for: **labial adhesions** (fused labia minora with a thin midline line), **clitoromegaly** (CAH), discharge (vulvovaginitis: poor hygiene, threadworms, foreign body; gonorrhoea or trichomonas = abuse until proven otherwise), bleeding, erythema, **injury**, warts, an imperforate hymen (bulging bluish membrane in a pubertal girl with cyclical pain and amenorrhoea).
4. **Never** use a speculum or do a digital vaginal examination in a prepubertal girl on the ward.

M. Anus and perineum (inspection)

1. Gently part the buttocks: **fissure** (constipation), **rectal prolapse** (*Shigella*, *Trichuris*, cystic fibrosis), **threadworms**, perianal excoriation, warts, abscess, and **injury** (abuse); in a newborn, **patency and position** of the anus (imperforate anus, fistula).
2. **Digital rectal examination**: only if specifically indicated (for example suspected Hirschsprung disease, intussusception, pelvic abscess), by or with a senior, using the **little finger** in infants.

N. Completing the examination

1. **Urine dipstick** (and microscopy and culture if abnormal) on a clean-catch or bag specimen.
2. **Blood pressure** with a correctly sized cuff (if not done in step B), compared with age and height centiles.
3. **Plot growth**; **Tanner stage** in adolescents.
4. Examine **other systems** as relevant: chest (lower-lobe pneumonia), heart (murmurs, failure), skin, joints; **ears and throat last** (distressing).
5. Cover the child, **thank** the child and parent, **wash hands**.
6. **Explain the findings** to the parent and child and **summarise** (section 9).

6. Findings and Interpretation

Inspection and palpation signs

Finding	What it means
Child lying still, abdomen not moving with breathing	Peritonitis
Writhing	Colic (intussusception, renal colic, obstruction)
Scaphoid abdomen	Congenital diaphragmatic hernia (newborn); dehydration
Distension (5 Fs + organomegaly or masses)	Tense, shiny, with dilated veins or fluid = true distension; a soft pot belly in a toddler is normal
Visible peristalsis LUQ → right after a test feed	Pyloric stenosis
Dilated veins, flow away from the umbilicus	Portal hypertension (schistosomiasis, portal vein thrombosis)
Dilated veins, flow upward	IVC obstruction
Everted umbilicus	Ascites
Bruising on the abdomen	Trauma or abuse
Involuntary guarding, rigidity, pain on cough or hop	Peritonism (perforated appendix, typhoid perforation)
Indentable mass in the left iliac fossa	Faeces (constipation)
Suprapubic dull mass you cannot get below, disappears after voiding	Full bladder; persistent in a boy with a poor stream = posterior urethral valves
Liver edge > 2 cm, firm, nodular or rounded, with an increased span	True hepatomegaly
Liver edge low but span normal	Displaced by hyperinflation (bronchiolitis, asthma)

Finding	What it means
Tender smooth liver + raised JVP, gallop, oedema	Heart failure
Pulsatile liver	Tricuspid regurgitation
Palpable spleen beyond infancy	Splenomegaly until proven otherwise
Shifting dullness, fluid thrill (tense)	Ascites
Absent bowel sounds	Ileus, peritonitis
High-pitched, tinkling bowel sounds	Obstruction
Succussion splash	Gastric outlet obstruction
Renal bruit in a hypertensive child	Renal artery stenosis
Weak femorals, radio-femoral delay	Coarctation of the aorta

Spleen or left kidney?

Feature	Spleen	Kidney
Notch	Present (medial border)	Absent
Movement with inspiration	Down and **toward the right iliac fossa**	Slight, downward
Get above it?	**No**	**Yes**
Ballotable?	No	**Yes**
Percussion	**Dull**	**Resonant band** (colon in front)

Abdominal masses

Feature	Wilms tumour	Neuroblastoma	Burkitt lymphoma
Age	2–5 years	Often under 2	Often school age (for example 6 years)
Child	Well	Unwell	Rapid growth; may be unwell
Mass	Smooth flank mass, **does not cross the midline**, ballotable	Irregular, hard, **crosses the midline**, calcification	Rapidly enlarging intra-abdominal mass, ± ascites
Associated	± Haematuria, hypertension	Raccoon eyes, bone pain, hypertension	**Jaw or orbital swelling**; tumour lysis risk

Groin and scrotal swellings

Feature	Inguinal hernia	Hydrocele
Get above it?	**No**	**Yes**
Cough or cry impulse	Yes	No
Reducible	Usually	No
Transilluminates	Usually not (infant bowel can transmit some light)	**Yes**
Testis	Separate, palpable	May be impalpable within it

Acute scrotum	Key signs
Testicular torsion	High-riding, **horizontal** testis, tender, **absent cremasteric reflex**; red or blue scrotum

Acute scrotum	Key signs
Torsion of the appendix testis	**Blue dot** and tenderness at the upper pole; cremasteric reflex present
Epididymo-orchitis	Tender epididymis; urine abnormal; mumps orchitis 4–6 days after parotitis
Incarcerated hernia	Irreducible, tender groin swelling, vomiting, distension

Genital findings

Finding	What it means
Ventral meatus, dorsal hooded foreskin, chordee	Hypospadias (with an undescended testis: think DSD or CAH)
Non-retractile foreskin, ballooning on voiding	Normal
Scarred white ring at the tip of the foreskin	True phimosis (balanitis xerotica obliterans)
Underdeveloped rugae on one side	Undescended testis on that side
Bilateral impalpable testes in a newborn, or clitoromegaly	CAH or DSD until proven otherwise
Fused labia minora with a thin midline line	Labial adhesions
Gonorrhoea or trichomonas discharge, genital injury in a prepubertal child	Sexual abuse until proven otherwise
Bulging bluish membrane, cyclical pain, amenorrhoea	Imperforate hymen
Fissure, rectal prolapse, threadworms	Constipation; *Shigella*, *Trichuris*, cystic fibrosis; threadworm infection

Bedside tests that complete the examination

Test	How	What it tells you
Urine dipstick	Fresh clean-catch or bag sample	**Blood** (AGN, UTI, Wilms, *S. haematobium*, trauma), **protein** (nephrotic: 3–4+), **nitrites and leucocytes** (UTI), **glucose and ketones** (DKA), **bilirubin and urobilinogen** (liver versus haemolysis)
Blood pressure	Correct cuff; compare with centiles	Hypertension (AGN, renal disease, Wilms, neuroblastoma, coarctation)
Malaria RDT	Finger prick	Malaria with splenomegaly
HemoCue Hb	Finger prick	Anaemia (SCD, malaria, leukaemia, sequestration)
Sickling test / HbSS point-of-care test	Where available	SCD (splenomegaly in infancy, crises)
Random blood glucose	Glucometer	DKA presenting as abdominal pain
Transillumination	Torch against the scrotal swelling in a dark room	Hydrocele versus hernia or tumour
Abdominal girth	Tape at the umbilicus, daily	Ascites trend
Bedside ultrasound (POCUS), if skilled	Probe on the abdomen	Free fluid, hydronephrosis, intussusception (target sign), bladder volume

Investigations prompted by the findings

Test	When	Expected or abnormal finding	How it changes management
Abdominal ultrasound	Mass, organomegaly, ascites, suspected intussusception, hydronephrosis; undescended testis location (limited value)	Organ of origin, solid versus cystic, periportal fibrosis (schistosomiasis), target sign	Biopsy, surgery, air or saline reduction. Most HC IVs (if a sonographer) and all hospitals
Scrotal Doppler ultrasound	Only if torsion is **unlikely** after surgical review	Absent flow (torsion), increased flow (epididymo-orchitis)	**Never delay surgery** for imaging when torsion is likely (RCH)
FBC and blood film	Hepatosplenomegaly, pallor, bruising	Blasts (leukaemia), sickle cells, malaria parasites, eosinophilia (schistosomiasis), pancytopenia (hypersplenism, leishmaniasis)	Specific treatment; bone marrow at RRH or NRH
LFTs, albumin	Hepatomegaly, jaundice, ascites, oedema	Raised transaminases, low albumin	Hepatitis versus nephrotic syndrome versus malnutrition
Urinalysis, urine culture, urine protein:creatinine ratio	Abnormal dipstick	Infection, nephrotic-range proteinuria	Antibiotics; steroids (nephrotic)
Urea, creatinine, electrolytes	Hypertension, haematuria, AKI, ambiguous genitalia	Raised creatinine; low Na and high K in CAH	AKI care; hydrocortisone
Stool microscopy (Kato-Katz), rectal snip	Hepatosplenomegaly in a lakeside child	*S. mansoni* eggs	**Praziquantel 40 mg/kg** single dose
rK39 rapid test, splenic or bone marrow aspirate	Fever, massive spleen, wasting in Karamoja or the north-east	Visceral leishmaniasis	Treatment at Amudat, Kacheliba or Moroto centres
HIV test, Xpert MTB, ascitic tap	Ascites, lymphadenopathy, wasting	HIV, TB peritonitis (exudate, lymphocytes)	ART, TB treatment
Fine-needle aspirate or biopsy	Abdominal mass, jaw mass	**Burkitt lymphoma** (starry sky), Wilms	Chemotherapy at the **Uganda Cancer Institute**; avoid FNA of suspected Wilms (biopsy decisions by surgeons and oncologists)
Echocardiogram	Hepatomegaly with heart failure, or ascites with a raised JVP	RHD, cardiomyopathy, **EMF**, pericardial effusion	Cardiac treatment (Uganda Heart Institute)
Karyotype, 17-OH-progesterone	Ambiguous genitalia, bilateral impalpable testes	CAH, DSD	Specialist referral

7. Clinical Acumen

Red flags

RED FLAG

- **Bilious (green) vomiting** in an infant: malrotation with volvulus until proven otherwise (surgical emergency).
- **Acute scrotal pain or swelling**: torsion until proven otherwise; theatre within 6 hours.
- **Irreducible, tender groin swelling**, vomiting, distension: incarcerated or strangulated hernia.
- **Abdominal mass** in a child (Burkitt, Wilms, neuroblastoma): urgent referral; palpate gently.
- **Rigid, silent abdomen**, guarding, rebound: peritonitis (perforated appendix, typhoid perforation).
- **Massive splenomegaly with sudden pallor and shock** in a child with SCD: **splenic sequestration** (transfuse now).
- **Hypertension with haematuria and oedema**: AGN with risk of encephalopathy.
- **Genital injury, bleeding or discharge in a prepubertal child**: possible sexual abuse.
- **Ambiguous genitalia with vomiting in a newborn**: salt-wasting CAH (check Na, K, glucose).

Thinking like a clinician

- **Pain in the abdomen may come from elsewhere**: lower-lobe pneumonia, testicular torsion, DKA, sickle cell crisis, HSP, UTI, otitis and tonsillitis (mesenteric adenitis), and constipation.
- **A big liver is not always a big liver**: hyperinflated lungs (bronchiolitis, asthma) push it down; measure the span.
- **Tender, smooth hepatomegaly with a raised JVP, gallop and oedema = heart failure** (rheumatic heart disease, cardiomyopathy, severe anaemia).
- **Massive splenomegaly** in Uganda: hyperreactive malarial splenomegaly, schistosomiasis with portal hypertension, visceral leishmaniasis (Karamoja), chronic myeloid leukaemia (rare), and **SCD** (classically the spleen auto-infarcts by school age, but in malaria-endemic Uganda **persistent splenomegaly** in older children with SCD is common).
- **Ascites with a normal liver and heavy proteinuria = nephrotic syndrome**; with a **raised JVP and a small heart sound** think **endomyocardial fibrosis** (EMF, still seen in Uganda) or constrictive pericarditis (TB); with fever and weight loss think **TB peritonitis**.
- **Abdominal mass + jaw swelling in a 6-year-old** in Uganda = **Burkitt lymphoma** (fast-growing; tumour lysis risk).
- **A "hernia" that you can get above and that transilluminates is a hydrocele**; a scrotal swelling that you cannot get above is a hernia.

Judgement calls

- **Crying child**: stop, play, use the mother's lap, palpate between cries during inspiration, while feeding or asleep, or after analgesia. A useless examination is worse than a delayed one.
- **When to stop and escalate**: emergency signs (ETAT), a suspected tumour (one gentle examination, then ultrasound and referral), suspected torsion (surgeon now, no ultrasound first), suspected abuse (stop unnecessary examination, call the senior).

Common student mistakes

- Starting palpation where it hurts.
- Examining the abdomen standing above a crying child.

- Forgetting the groins and testes.
- Forcibly retracting a foreskin.
- Calling a normal infant liver "hepatomegaly".
- Missing a full bladder (the "mass" that disappears after voiding).
- Not measuring BP; not dipping the urine.

Worked examples

EXAM FAVOURITE

Worked example 1: spleen or kidney? A 4-year-old has a left upper quadrant mass. It **moves down and toward the right iliac fossa** with inspiration, you **cannot get above it**, there is a **notch** on its medial border, it is **not ballotable**, and it is **dull** to percussion → **spleen**. If instead the mass were **ballotable**, you **could get above it**, it moved only slightly downward with breathing and had a **resonant band** over it (colon in front) → **kidney** (Wilms tumour, hydronephrosis).

EXAM FAVOURITE

Worked example 2: is this liver really big? A 5-month-old with bronchiolitis has a liver edge **3 cm** below the costal margin. Percussion: upper border of dullness at the **7th intercostal space** (low because of hyperinflation), span **5 cm** (normal for age). Interpretation: **displaced, not enlarged** liver. Contrast: a 5-month-old with a **tender liver 3 cm**, gallop, and RR 70 → **heart failure** (large VSD), where the liver span is increased and the edge rounded.

8. Complications and Troubleshooting

Complications of the examination

Situation	Risk	Prevention
Vigorous palpation of Wilms or neuroblastoma	Tumour rupture and spread	Gentle, single examiner
Forcible foreskin retraction	Tears, scarring, **pathological phimosis**, paraphimosis	Never retract
Missed testicular torsion ("tummy ache")	Loss of the testis	Examine the scrotum in every boy with abdominal pain
Missed incarcerated hernia	Bowel strangulation, testicular infarction	Examine the groins in every vomiting infant
Circumcision of a hypospadic boy	Loss of tissue for repair	Examine the meatus first
Rough or unexplained genital examination	Psychological harm; retraumatisation in abuse	Consent and assent, chaperone, inspection only
Over-diagnosis of hepatomegaly	Unnecessary tests	Measure the span
Cold hands, standing over the child	Crying, guarding, a useless examination	Warm hands, child's eye level, play

Managing what you find: organomegaly

Finding	Likely causes (Uganda)	First steps
Splenomegaly ± hepatomegaly with fever	**Malaria**	RDT or smear; treat (ACT or artesunate per severity)

Finding	Likely causes (Uganda)	First steps
Massive splenomegaly, recurrent malaria, high IgM	**Hyperreactive malarial splenomegaly**	Long-term antimalarial prophylaxis per specialist; exclude lymphoma
Firm hepatosplenomegaly, lakeside, portal hypertension	**Schistosomiasis (*S. mansoni*)**	Stool Kato-Katz, ultrasound (periportal fibrosis); **praziquantel 40 mg/kg single dose**; watch for variceal bleeding
Splenomegaly in an infant or young child with pallor, dactylitis, jaundice	**Sickle cell disease**	Sickling or HbS test; folic acid, penicillin V prophylaxis, pneumococcal vaccine, malaria prevention; teach parents to **palpate the spleen daily** (sequestration)
Acute splenic enlargement + sudden pallor + shock in SCD	**Splenic sequestration**	**Emergency transfusion** (10 mL/kg packed cells, cautiously), fluids, senior
Massive spleen, fever, wasting, pancytopenia (Karamoja, north-east)	**Visceral leishmaniasis (kala-azar)**	rK39, refer to a treatment centre
Hepatosplenomegaly + lymphadenopathy + bruising or bone pain	**Leukaemia**, lymphoma	FBC and film, urgent referral (Uganda Cancer Institute)
Hepatosplenomegaly + thrush, parotids, failure to thrive	**HIV** (± TB, CMV)	HIV test or PCR, ART, TB screen
Tender smooth hepatomegaly, raised JVP, gallop, oedema	**Heart failure** (RHD, cardiomyopathy, severe anaemia, large shunts)	Oxygen, furosemide **1 mg/kg**, treat the cause; echo
Tender hepatomegaly with fever	Amoebic or pyogenic liver abscess, viral hepatitis, typhoid	Ultrasound; metronidazole for amoebic abscess

Managing what you find: abdominal masses

1. **Do not over-palpate** (tumour rupture); **one examiner**, gently.
2. **Measure BP** (Wilms and neuroblastoma can cause hypertension), dip the urine (haematuria in Wilms), check for **jaw or orbital masses** (Burkitt) and raccoon eyes (neuroblastoma).
3. **Ultrasound** first; then refer urgently.
4. **Burkitt lymphoma**: very fast growing; before chemotherapy start **hydration and allopurinol** (tumour lysis prevention); treatment at the **Uganda Cancer Institute** or regional centres, with an excellent response to chemotherapy.
5. **Faecal masses** (indentable, left iliac fossa, constipation history): disimpaction and laxatives (lactulose; see constipation guidance).
6. **Full bladder**: re-examine after voiding; a persistently distended bladder in a boy → **posterior urethral valves** (catheterise, urology).

Managing what you find: ascites

- Confirm (shifting dullness, girth), then look for the cause: **urine protein** (nephrotic syndrome → steroids per the nephrotic syndrome topic), **JVP, heart sounds, echo** (EMF, constrictive pericarditis, heart failure), **liver and spleen** (portal hypertension), **TB screen** (ascitic tap: exudate, lymphocytes; Xpert), **albumin** (malnutrition, protein-losing states).
- Diagnostic **ascitic tap** by a senior, with ultrasound guidance where available.

Managing what you find: groin and scrotum

Finding	Management
Reducible inguinal hernia	**Refer for elective herniotomy soon** (within weeks; sooner in infants and preterms because of the incarceration risk). Teach parents the signs of incarceration
Incarcerated (irreducible) hernia, no signs of strangulation	Analgesia, position, **gentle sustained manual reduction (taxis)** by a senior; if reduced, herniotomy within 24–48 h; if not reducible → emergency surgery
Strangulated hernia (tender, red, vomiting, distension, unwell)	**Nil by mouth, IV fluids, analgesia, emergency surgery**
Hydrocele (transilluminates, can get above, testis normal)	**Observe**: most close by **1–2 years**; refer if it persists beyond 1–2 years, fluctuates in size (communicating), or is very large
Testicular torsion suspected	**Emergency**: analgesia, **nil by mouth**, **urgent surgical review and exploration** (do not delay for ultrasound); viability 80–100% if operated within **4–6 hours**; contralateral orchidopexy
Torsion of the appendix testis (blue dot, upper-pole tenderness, cremasteric reflex present)	Analgesia, rest, scrotal support; settles in about a week (if any doubt, explore)
Epididymo-orchitis	Urinalysis and culture; antibiotics per organism; in infants look for a urinary tract anomaly; in adolescents consider STI and safeguarding; mumps orchitis 4–6 days after parotitis
Undescended testis	Re-examine at **6–8 weeks** and **3 months**; refer if not in the scrotum by **6 months** for **orchidopexy at 6–12 months (by 18 months)**; bilateral impalpable testes in a newborn → urgent senior review (exclude **CAH or DSD**: electrolytes, karyotype)
Retractile testis	Reassure; review yearly until puberty (may "ascend")
Umbilical hernia	Reassure; most close by **4–5 years**; refer if persisting beyond about 4–5 years, a very large defect, or ever incarcerated; no coins or binders

Managing what you find: penis and foreskin

Finding	Management
Physiological non-retractile foreskin	Reassure; **never forcibly retract**; clean the outside only
Ballooning on voiding	Normal; topical **0.05% betamethasone** to the preputial ring 3 times daily for 6–12 weeks only if problematic (RCH)
Balanitis	Warm salt-water soaks, barrier cream or 1% hydrocortisone, antifungal if candida; antibiotics only if cellulitis
True (pathological) phimosis (scarred, white, fibrotic ring; BXO)	Topical steroid trial; **circumcision** by surgeons if scarred or recurrent infections
Paraphimosis	**Emergency**: analgesia (local anaesthetic gel, penile block), compress the oedematous glans, and reduce the foreskin forward; surgical help if this fails
Hypospadias	**Do not circumcise** (the foreskin is used in the repair); refer to paediatric surgery or urology (repair usually **6–18 months**); if hypospadias with an **undescended testis**, check for **DSD or CAH**
Micropenis or ambiguous genitalia	Do not assign sex in a hurry; check **electrolytes and glucose** (salt-wasting CAH); urgent specialist referral

UGANDA CONTEXT

Circumcision is common in Uganda (Muslim families, cultural ceremonies such as *imbalu* among the Bagisu, and voluntary medical male circumcision for HIV prevention in adolescents). Before any neonatal or infant circumcision, **examine for hypospadias and chordee** and make sure both testes are descended. Many families seek surgery for a normal non-retractile foreskin; reassurance is the right treatment. Paediatric surgical services are at **Mulago, Mbarara, Gulu, Mengo, CoRSU** and other referral centres.

Managing what you find: female genitalia

Finding	Management
Neonatal discharge or withdrawal bleed	Reassure (maternal oestrogen)
Labial adhesions	Reassure if asymptomatic (most separate at puberty); **topical oestrogen cream** for 2–4 weeks if urinary symptoms; never forcibly separate
Vulvovaginitis (poor hygiene, threadworms, irritants)	Hygiene advice, cotton underwear, avoid soaps and bubble baths, treat threadworms (**albendazole 400 mg**, or 200 mg if under 2 years); persistent bloody discharge → foreign body, refer
Clitoromegaly	Senior review the same day; electrolytes (**CAH**)
Imperforate hymen (pubertal girl, cyclical pain, bulging bluish membrane)	Gynaecology referral for hymenotomy
Suspected sexual abuse	**Safety first**: inform the senior; **child protection** (probation and social welfare officer, police; Police Form 3 completed by an authorised medical officer); **HIV PEP within 72 hours**, emergency contraception (pubertal girls) within 120 h, STI prophylaxis, hepatitis B vaccination; forensic examination by a trained clinician; document exact words and findings; psychosocial support

Drug doses used in this topic

Drug	Dose	Notes
Paracetamol	**15 mg/kg** 6-hourly (max 60 mg/kg/day; max 1 g/dose), oral or IV	Analgesia does not mask surgical signs
Morphine	**0.1 mg/kg IV**, titrated	Severe pain; by a senior
Praziquantel	**40 mg/kg** single dose	*S. mansoni*
Furosemide	**1 mg/kg**	Heart failure
Packed cells	**10 mL/kg**, cautiously	Splenic sequestration
Betamethasone 0.05%	To the preputial ring 3 times daily for **6–12 weeks**	Problematic ballooning, phimosis (RCH)
Albendazole	**400 mg** (200 mg if under 2 years)	Threadworms

9. Documentation and Presenting Findings

Document these positives and negatives

Document these positives	And these negatives
"Liver 4 cm below RCM, span 9 cm, firm, non-tender, smooth"	"No ascites (no shifting dullness)"

Document these positives	And these negatives
"Spleen 6 cm along its axis, Hackett grade 3, firm, notch felt"	"Kidneys not ballotable"
"10 × 8 cm firm, smooth mass in the left flank, does not cross the midline, ballotable"	"Both testes descended, normal lie, non-tender"
"Right inguinal swelling, cannot get above, reducible, cough impulse"	"Foreskin not retracted (normal)"; "No hypospadias"
BP and centile; urine dipstick result	"No lymphadenopathy, no jaundice"

Record the **chaperone's name**.

Model one-line summary

> "A 7-year-old boy from Buliisa (Lake Albert) with 6 months of abdominal swelling and occasional blood in the stool: well-grown (WAZ –1.2), pale, **firm hepatomegaly (left lobe prominent) 5 cm** below the costal margin with a **span of 11 cm**, **firm splenomegaly 8 cm (Hackett grade 3)**, dilated periumbilical veins, no ascites, BP normal, urine dipstick negative: **hepatosplenomegaly with signs of portal hypertension, most likely hepatosplenic schistosomiasis (*S. mansoni*)**; differentials: hyperreactive malarial splenomegaly, chronic liver disease, haematological malignancy."

Model case presentation (ward round)

> "Opio is a **6-year-old boy from Lira**, weight **17 kg (WAZ –1.6)**, height 112 cm (HAZ –0.8), **fully immunised** on UNEPI, **HIV-negative** (tested this admission), eating family foods (posho, beans, greens; little animal protein), with a **3-week history of a rapidly enlarging abdomen**, reduced appetite, night sweats and weight loss. No jaw swelling, no haematuria. On examination he is **pale**, not jaundiced, with no lymphadenopathy. The abdomen is **distended and asymmetrical**. There is a **14 × 10 cm hard, irregular, non-tender mass in the right iliac fossa and periumbilical region**, which **does not move with respiration**, is **dull** to percussion, and **crosses the midline**; I can get above it. The liver and spleen are not separately palpable; kidneys not ballotable. **Shifting dullness present** (moderate ascites). Bowel sounds normal. **Both testes descended**, no hernias. **BP 104/66** (normal for age). **Urine dipstick: no blood or protein. Summary**: a rapidly growing intra-abdominal mass with ascites in a 6-year-old in northern Uganda: **abdominal Burkitt lymphoma** is most likely; differentials include neuroblastoma, abdominal TB and Wilms tumour (less likely: crosses the midline, no haematuria). **Plan**: ultrasound, FBC, LDH, uric acid, creatinine and electrolytes, ascitic fluid cytology or FNA of the mass, **IV hydration and allopurinol** (tumour lysis prevention), and urgent referral to the Uganda Cancer Institute or Lacor paediatric oncology."

After the examination

1. Present findings with **relevant positives and negatives**.
2. Request the **right next test** (ultrasound, FBC, urinalysis, specific tests) and the **right referral** (surgery, oncology, cardiology, nephrology).
3. **Explain** to the parent and child, and agree a plan and safety-netting (for example hernia incarceration signs, daily spleen palpation in SCD).
4. Document clearly, including the **chaperone's name**.

10. OSCE Checklist

Step	Done well when	Critical?
1. Introduction	Introduces self to parent and child; confirms name and age; explains in simple language	Yes

Step	Done well when	Critical?
2. Consent and chaperone	Consent (parent) and assent (child); **offers or uses a chaperone** for the genital examination	Yes
3. Hand hygiene	Washes and **warms hands**	Yes
4. Positioning	Positions for age (mother's lap or supine, knees flexed); **gets to the child's eye level**; uses distraction	No
5. Exposure	Exposes appropriately, preserving dignity	No
6. General examination	Well or unwell, **growth and MUAC**, hands, pallor, jaundice, mouth, nodes, oedema, skin	No
7. Inspection	Shape, movement with breathing, veins, scars, umbilicus, visible peristalsis or masses, groins	No
8. Pain first	Asks about pain; **starts away from the painful area**; **watches the face**	Yes
9. Palpation	**Light then deep** palpation of all regions	No
10. Liver	Palpated from the RIF upward; describes size in cm and characteristics	No
11. Spleen	Palpated from the RIF toward the LUQ; rolls the child to the right if not felt	No
12. Kidneys and bladder	Kidneys ballotted bimanually; bladder assessed	No
13. Percussion	**Percusses liver span**; checks for **shifting dullness**	No
14. Auscultation	Bowel sounds (± bruits)	No
15. Groins	Examines for hernias; feels **femoral pulses**	No
16. Genitalia	**Male**: inspects penis (meatus, foreskin **not retracted**), scrotum; palpates testes after milking down from the groin; transilluminates a swelling. **Female**: inspection only, frog-leg position	Yes
17. Perianal	Inspects; states DRE is not routine	No
18. Completion	States **BP, urine dipstick, plot growth**, examine other systems	Yes
19. Closure	Covers the child, thanks parent and child, washes hands	No
20. Presentation	**Presents findings** clearly with positives, relevant negatives and a differential	Yes

ll. Key Points

High-yield points

- **Watch the face, warm the hands, start far away** from the pain.
- General examination and growth first; finish with **BP, urine dipstick and growth plotted**.
- A liver edge up to **1–2 cm** is normal in infants; **measure the span**.
- Start liver and spleen palpation in the **right iliac fossa**.
- Spleen: notch, moves toward the RIF, cannot get above, dull. Kidney: ballotable, can get above, resonant.
- **Every boy with abdominal pain gets his testes examined**; torsion = theatre within 4–6 hours, no ultrasound first.
- Hernia: cannot get above. Hydrocele: can get above and transilluminates.
- **Never** forcibly retract a foreskin; **never** circumcise a boy with hypospadias.

- Prepubertal girls: **inspection only**, with a chaperone.
- Suspected Wilms or neuroblastoma: **one gentle examination**.

Numbers & doses

Item	Value
Normal liver edge	Up to **1–2 cm** below the RCM in infants and toddlers (up to ~3 cm in newborns), soft and sharp
Liver span	About **4.5–5 cm** at 1 week; about **7–8 cm (boys) / 6–6.5 cm (girls)** at 12 years
Spleen	Tip may be felt in healthy neonates and young infants; otherwise abnormal
Hackett grades	**0–5** (costal margin → halfway to umbilicus → umbilicus → halfway to pubis → beyond)
Fast breathing (context)	≥ 60 (< 2 mo), ≥ 50 (2–12 mo), ≥ 40 (1–5 y)
BP cuff	Bladder **80–100%** of arm circumference, width **≥ 40%**
Undescended testis	Refer if not down by **6 months**; orchidopexy **6–12 months** (≤ 18 months)
Testicular torsion	Theatre within **4–6 h**: 80–100% salvage
Hydrocele	Usually resolves by **1–2 years**
Umbilical hernia	Usually closes by **4–5 years**
Foreskin retractable	**10%** at 1 y, **50%** at 10 y, **99%** at 17 y
Ballooning or phimosis steroid	**Betamethasone 0.05%** to the preputial ring 3 times daily for **6–12 weeks** (RCH)
Micropenis	Stretched length **< 2.5 cm** at term
Puberty onset	Girls **8–13 y** (breast bud); boys **9–14 y** (testes ≥ **4 mL**)
Precocious / delayed	< **8** / > **13** (girls); < **9** / > **14** (boys)
Paracetamol	**15 mg/kg** 6-hourly (max 60 mg/kg/day; max 1 g/dose)
Praziquantel (schistosomiasis)	**40 mg/kg** single dose
Sexual abuse	HIV **PEP within 72 h**; emergency contraception within **120 h**

Exam traps

- **"Palpable liver edge in an infant = hepatomegaly."** Wrong: up to 1–2 cm is normal; measure the span.
- **"Start palpating at the costal margin."** Wrong: start in the **RIF** for liver and spleen, or you will miss a big organ.
- **"Retract the foreskin to clean it."** Wrong: never forcibly retract.
- **"Circumcise a boy with a hooded foreskin and ventral meatus."** Wrong: hypospadias; refer first.
- **"Order an ultrasound before surgery for suspected torsion."** Wrong: explore urgently.
- **"Scrotal swelling transilluminates, so it is a hernia."** Wrong: transillumination suggests hydrocele.
- **"Examine a prepubertal girl with a speculum for discharge."** Wrong: inspection only; refer.
- **"Abdominal pain, so the cause is in the abdomen."** Wrong: check the testes, the chest (lower-lobe pneumonia), glucose (DKA) and the throat; and in SCD think vaso-occlusive crisis or sequestration.

Questions seniors ask

Viva questions

1. *How do you distinguish spleen from left kidney?* Spleen: notch, moves toward the RIF with inspiration, cannot get above it, dull to percussion, not ballotable. Kidney: ballotable, can get above it, resonant band over it, moves little.
2. *How do you tell a hernia from a hydrocele?* Hernia: cannot get above it, cough or cry impulse, reducible, usually not transilluminant. Hydrocele: can get above it, transilluminates, no impulse, testis may be within it.
3. *What is a retractile testis?* A descended testis pulled up by an active cremasteric reflex that can be brought down to the scrotum and stays; normal; review yearly.
4. *When should an undescended testis be operated on?* Refer if not down by 6 months; orchidopexy at 6–12 months, by 18 months at the latest (fertility and cancer risk).
5. *Why should you not circumcise a boy with hypospadias?* The foreskin is used for the reconstruction.
6. *Name causes of hepatosplenomegaly in Ugandan children.* Malaria (and hyperreactive malarial splenomegaly), schistosomiasis, SCD (early), HIV, TB, leukaemia or lymphoma, visceral leishmaniasis, heart failure (liver), typhoid.
7. *What is the Hackett grading?* A 0–5 scale of spleen size relative to the costal margin, umbilicus and pubic symphysis, used in malaria surveys.
8. *How do you assess for ascites?* Inspection (distension, flanks, everted umbilicus), shifting dullness, fluid thrill (tense ascites), girth; then look for the cause (urine protein, JVP and heart, liver, TB).

On the ward round

- **How would you examine the abdomen of a crying 18-month-old?** On the mother's lap, warm hands, distraction (toy, bubbles), palpate during inspiration between cries or while the child is feeding or asleep, palpate with the child's hand under mine, and watch the face for tenderness.
- **This boy has abdominal pain. What part of the examination must you never skip?** The groins and testes (torsion and incarcerated hernia), plus the chest (lower-lobe pneumonia), a urine dipstick and glucose.
- **How do you describe a mass?** Site, size, shape, surface, edge, consistency, tenderness, mobility (with respiration and side to side), whether you can get above or below it, whether it crosses the midline, percussion note, pulsatility, bruit, and transillumination; then BP and urine.
- **What features distinguish Wilms tumour from neuroblastoma?** Wilms: younger child (2–5 y), well, smooth flank mass not crossing the midline, ± haematuria and hypertension. Neuroblastoma: often under 2, unwell, irregular hard mass crossing the midline, calcification, raccoon eyes, bone pain.
- **A testis is not in the scrotum. How do you decide if it is retractile or undescended?** Examine warm and relaxed (cross-legged), milk from the inguinal canal downward; if it can be brought to the bottom of the scrotum and stays briefly, it is retractile; if not, or it springs back immediately, it is undescended; if not felt anywhere, it is impalpable (needs surgical or laparoscopic assessment).
- **What is your duty if you suspect sexual abuse during an examination?** Stop unnecessary examination, ensure the child's safety, inform the senior, document exactly, start time-sensitive care (PEP within 72 h, emergency contraception, STI prophylaxis), and involve child protection and police according to Ugandan procedures; the forensic examination is done once by a trained examiner.

Mnemonics

- **"Watch the face, warm the hands, start far away."**
- **Distension "5 Fs + O"**: Fluid, Flatus, Faeces, Fat, Fetus, plus **O**rganomegaly or masses.
- **Spleen versus kidney "SNAP-DB"**: Spleen has a **N**otch, moves **A**cross toward the RIF, no **P**ossibility of getting above; **D**ull; kidney is **B**allotable.
- **Scrotal swelling "Above, Light, Separate"**: can you get **above** it? Does it **transilluminate**? Is the testis **separate**?
- **Torsion: "6 hours to save the testis."**
- **Foreskin "10–50–99"**: retractable in 10% at 1 year, 50% at 10 years, 99% at 17 years.

- **Wilms versus neuroblastoma**: **W**ilms is **W**ell with a **W**ithin-one-side mass; **N**euroblastoma is **N**ot well and **N**ot respecting the midline.
- **Puberty "8–9 too early, 13–14 too late"** (girls/boys).

OSCE tips

- Start with **introduction, consent and assent, and a chaperone**.
- **Warm hands and kneel**.
- **Say what you are doing** while you do it ("I am starting away from where it hurts and watching his face").
- Do the **general examination and growth first**.
- Examine **liver and spleen from the right iliac fossa**.
- **Examine the groins and genitalia** (or state that you would, with a chaperone, if the station uses a manikin); **do not retract the foreskin**.
- Finish with **"To complete my examination I would check the blood pressure, dip the urine and plot his growth."**

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SELF-TEST

Flashcards

29 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Sequence of the paediatric abdominal and GU examination	Introduce, **consent + assent**, **chaperone**, hand hygiene, **warm hands**, position for age, expose (dignity), **general exam and growth**, inspect, palpate (light → deep), percuss, auscultate, **groins, genitalia, perianal inspection**, then **BP, urine dipstick, plot growth**.
2	How to position a toddler for abdominal examination	On the **mother's lap** (or supine with the mother holding hands at the head end), **knees flexed**, examiner **at the child's eye level**, with **distraction** (toys, bubbles).
3	Where do you start palpating, and what do you watch?	Start in the quadrant **farthest from the pain**; **watch the child's face** (grimace, pause in crying) rather than your hand.
4	Normal liver edge in infants and young children	Soft, sharp edge up to **1–2 cm below the right costal margin** (up to ~3 cm in newborns). Confirm enlargement by **percussing the span**.
5	Why start liver and spleen palpation in the right iliac fossa?	A **very large organ** may extend into the lower abdomen; starting at the costal margin you would be already **on top of it** and miss the edge.
6	How to palpate the spleen if not felt supine	Roll the child **toward you onto the right side**, support the left lower ribs with your left hand, and palpate from below on inspiration.
7	Hackett grading of splenomegaly	**0** not palpable; **1** only on deep inspiration; **2** not beyond halfway costal margin–umbilicus; **3** not below umbilicus; **4** not beyond halfway umbilicus–pubis; **5** beyond that.
8	Spleen vs left kidney	**Spleen**: notch, moves toward the RIF on inspiration, **cannot get above**, dull, not ballotable. **Kidney**: **ballotable**, can get above, resonant band (colon) over it.
9	Describe an abdominal mass: checklist	Site, size, shape, surface, edge, consistency, tenderness, mobility (respiration, side to side), **get above/below?**, **crosses midline?**, percussion note, pulsation, bruit, transillumination.
10	Wilms tumour vs neuroblastoma	**Wilms**: well child 2–5 y, smooth flank mass **not crossing midline**, ± haematuria, hypertension. **Neuroblastoma**: often < 2 y, unwell, irregular hard mass **crossing midline**, raccoon eyes, bone pain.
11	Burkitt lymphoma clues (Uganda)	Rapidly growing **jaw or abdominal mass** (± ascites) in a 4–9-year-old; malaria/EBV belt; **tumour lysis** risk → hydration and allopurinol; very chemo-sensitive.
12	Causes of hepatosplenomegaly in Ugandan children	**Malaria** (and hyperreactive malarial splenomegaly), **schistosomiasis**, **SCD**, **HIV**, TB, **leukaemia/lymphoma**, **visceral leishmaniasis** (Karamoja), typhoid; liver alone: **heart failure**, hepatitis, abscess.
13	Shifting dullness: how?	Percuss from the midline to the flank until dull, keep the finger there, roll the child **away**, wait **30 s**, re-percuss: becomes **resonant** = free fluid.
14	Causes of ascites in Ugandan children	**Nephrotic syndrome**, heart failure, **endomyocardial fibrosis**, constrictive pericarditis, **TB peritonitis**, portal hypertension (schistosomiasis, cirrhosis), malnutrition, malignancy (Burkitt).
15	Visible peristalsis left to right in a 5-week-old with projectile vomiting	**Pyloric stenosis**: palpate the 'olive' in the RUQ/epigastrium during a test feed; check for hypochloraemic hypokalaemic alkalosis.

#	Question	Answer
16	Inguinal hernia vs hydrocele	**Hernia**: cannot get above, cough/cry impulse, reducible, usually not transilluminant. **Hydrocele**: **can get above**, **transilluminates**, no impulse.
17	Management of an inguinal hernia in an infant	**All need herniotomy**, promptly (high incarceration risk in infants); incarcerated → analgesia and gentle reduction by a senior, surgery in 24–48 h; strangulated → emergency surgery.
18	How to examine for undescended testes	Warm, relaxed child (cross-legged/squatting); **milk from the inguinal canal downward** before touching the scrotum; note position, size, whether it can be brought down and **stays** (retractile).
19	Timing of orchidopexy	Refer if not descended by **6 months**; operate at **6–12 months**, **by 18 months** at the latest.
20	Features of testicular torsion	Sudden severe pain ± abdominal pain and vomiting; **high-riding, horizontal** testis; **absent cremasteric reflex**; scrotal redness. **Explore within 4–6 h** (80–100% viability); no delay for ultrasound.
21	Blue dot sign	Small tender bluish nodule at the **upper pole** = **torsion of the appendix testis**; cremasteric reflex present; conservative management if certain.
22	Foreskin retractability with age	About **10% at 1 year**, **50% at 10 years**, **99% by 17 years**. Non-retractile foreskin is normal; **never forcibly retract**.
23	Hypospadias: features and key advice	**Ventral meatus**, **hooded foreskin**, **chordee**. **Do not circumcise** (foreskin needed for repair); refer for surgery (6–18 months); with undescended testis think **DSD/CAH**.
24	Paraphimosis	Retracted foreskin trapped behind the glans with swelling: **emergency** — analgesia, compress the glans, **reduce the foreskin forward**; surgical help if it fails.
25	Examining the genitalia of a prepubertal girl	**Inspection only**, frog-leg position or on mother's lap, **chaperone**, gentle labial separation outward and downward. **No speculum or digital exam.**
26	Persistently palpable bladder and poor urinary stream in an infant boy	**Posterior urethral valves**: catheterise, check creatinine and electrolytes, ultrasound, urology referral.
27	Tanner stage 2: first sign of puberty in boys and girls	**Boys**: testicular volume **≥ 4 mL** (9–14 y). **Girls**: **breast bud** (8–13 y). Precocious < 8 (girls) / < 9 (boys); delayed > 13 / > 14.
28	How to complete the abdominal exam (OSCE line)	'To complete my examination I would **measure the blood pressure**, **dip the urine**, **plot the growth**, and examine the chest, heart and other systems as relevant.'
29	Suspected sexual abuse found during examination	Stop unnecessary examination, ensure safety, inform the senior, document precisely, **HIV PEP within 72 h**, emergency contraception (pubertal girls), STI prophylaxis, child protection/police; forensic exam once by a trained clinician.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. You are asked to examine the abdomen of a crying 18-month-old with 2 days of fever and vomiting. He clings to his mother. What is the BEST approach?

- A Ask the nurse to hold him down on the couch so you can palpate quickly
- B Examine him on his mother's lap with warm hands, at his eye level, using distraction and watching his face
- C Skip the abdominal exam and order an ultrasound
- D Give a sedative before examining
- E Examine only when he is completely silent, even if it takes several hours

2. A 4-month-old with bronchiolitis has a liver edge 3 cm below the right costal margin. Percussion places the upper border of liver dullness at the 7th intercostal space and the span is 5 cm. The edge is soft and sharp. What is the MOST likely explanation?

- A Congestive heart failure
- B Hepatitis
- C A normal-sized liver displaced downward by hyperinflated lungs
- D Hepatoblastoma
- E Glycogen storage disease

3. A 5-year-old girl has a left upper quadrant mass. It moves toward the right iliac fossa on inspiration, you cannot get above it, a notch is felt on its medial border, it is not ballotable, and it is dull to percussion. What is the mass?

- A Left kidney
- B Spleen
- C Faecal mass in the descending colon
- D Left adrenal neuroblastoma
- E Left ovarian cyst

4. A 3-year-old boy is brought because his mother noticed a swelling in his abdomen while bathing him. He is well and playful. BP 118/78 (above the 95th centile). There is a smooth, firm 9 cm mass in the right flank that is ballotable and does not cross the midline. Urine dipstick: blood 2+. What is the MOST likely diagnosis?

- A Neuroblastoma
- B Burkitt lymphoma
- C Hydronephrosis from PUJ obstruction
- D Wilms tumour (nephroblastoma)
- E Splenomegaly

5. A 13-year-old boy presents with sudden severe left scrotal pain for 3 hours and vomiting. The left testis lies high with a horizontal lie and is very tender; the left cremasteric reflex is absent. What is the MOST appropriate next step?

- A Urgent surgical exploration (with analgesia and nil by mouth) without waiting for imaging
- B Scrotal Doppler ultrasound and review the result in the morning
- C Oral ciprofloxacin for epididymo-orchitis
- D Scrotal support and analgesia, review in 1 week
- E Urine culture and wait for the result

6. A mother brings her 9-month-old son because his left 'testicle is missing'. On examination in a warm room with the child sitting cross-legged, you milk the left inguinal canal downward and bring the testis to the bottom of the scrotum, where it stays briefly before rising again. The right testis is normal. What is the MOST appropriate management?

A Urgent orchidopexy
B Human chorionic gonadotrophin injections
C Refer for karyotyping
D Ultrasound and laparoscopy to find the testis
E Reassure: this is a retractile testis; review yearly

7. A 2-month-old boy has a right scrotal swelling that is larger in the evening, non-tender, and you cannot get above it. It is soft and reduces with gentle pressure. What is the MOST appropriate management?

A Observe until 2 years; most resolve
B Aspirate the fluid
C Apply a truss
D Refer for herniotomy soon (within weeks) and teach the parents the signs of incarceration
E Reassure; no follow-up needed

8. A 3-year-old boy's mother asks you to 'open' his foreskin because she cannot pull it back to clean. He passes urine normally with some ballooning; the foreskin tip is soft and unscarred. What is the BEST advice?

A Forcibly retract it in the clinic and apply petroleum jelly
B Book a circumcision for phimosis
C Dilate the preputial opening with forceps
D Start oral antibiotics
E Reassure: a non-retractile foreskin is normal at this age; clean the outside only and never force it back

9. A family in Mbale brings their 10-month-old son for traditional circumcision advice. You find the urethral meatus on the ventral surface of the glans, a hooded dorsal foreskin and mild ventral curvature. Both testes are descended. What is the MOST appropriate advice?

A Circumcise now; it will correct the problem
B Circumcise but leave the ventral skin
C No action is needed; this is a normal variant
D Do not circumcise; refer to paediatric surgery/urology for hypospadias repair, which uses the foreskin
E Refer for karyotyping before any discussion

10. A 7-year-old boy from Buliisa on Lake Albert has a firm liver 5 cm below the costal margin, a firm spleen 8 cm, dilated periumbilical veins, no ascites, and recurrent episodes of passing blood per rectum. Which investigation is MOST likely to confirm the diagnosis?

A Stool microscopy (Kato-Katz) for Schistosoma mansoni eggs and abdominal ultrasound for periportal fibrosis
B Blood film for sickle cells
C rK39 rapid test
D Bone marrow aspirate
E Urine microscopy for Schistosoma haematobium eggs

11. A 6-year-old girl has gross abdominal distension. Shifting dullness is present. The JVP is raised to the earlobes, heart sounds are soft, the liver is 6 cm and firm, and there is no peripheral oedema. Urine protein is negative. Albumin 33 g/L. Which diagnosis BEST fits? Advanced

A Nephrotic syndrome
B Kwashiorkor
C Endomyocardial fibrosis or constrictive pericarditis
D Burkitt lymphoma
E Cirrhosis from hepatitis B

12. A 4-year-old girl is brought with vaginal discharge and pain on passing urine. She is withdrawn and frightened. Which approach is MOST appropriate for examination?

A Speculum examination to obtain a high vaginal swab
B Digital vaginal examination to exclude a foreign body
C Examine under restraint to complete quickly
D Defer all examination and send home with antibiotics
E Inspect only, in the frog-leg position with the mother present and a chaperone, take a swab from the introitus, and involve a senior because of possible abuse

13. A 6-week-old boy has projectile, non-bilious vomiting after every feed for 1 week. He is hungry, mildly dehydrated, and has lost weight. During a test feed you see waves moving across the upper abdomen from left to right. Which finding on palpation would confirm the likely diagnosis?

A A sausage-shaped mass in the right upper quadrant
B A firm, mobile, olive-sized mass in the epigastrium/right upper quadrant
C A ballotable flank mass
D A distended, tympanic abdomen with absent bowel sounds
E A tender right iliac fossa

14. A newborn has bilateral impalpable testes and a small phallus with a ventral meatus. On day 8 he is feeding poorly and vomiting. What is the MOST important immediate investigation? Advanced

A Scrotal ultrasound
B Testosterone level
C Serum sodium, potassium and blood glucose
D Abdominal X-ray
E Urine culture

15. A 9-year-old boy with sickle cell disease presents with sudden pallor, lethargy and left upper quadrant fullness. HR 160, CRT 4 s. His spleen, previously 2 cm, is now 10 cm. Hb 3.8 g/dL (baseline 7.5). What is the MOST appropriate immediate management? Advanced

A Oral folic acid and review tomorrow
B Urgent blood transfusion and circulatory support for acute splenic sequestration
C Emergency splenectomy before any transfusion
D IV morphine and fluids for a vaso-occlusive crisis only
E Start antimalarials and wait for the blood smear

Answers & explanations

1. B · Examine him on his mother's lap with warm hands, at his eye level, using distraction and watching his face

B is correct: winning the child's cooperation (mother's lap, warm hands, distraction, eye level) gives a relaxed abdomen, and watching the face reveals tenderness. A is wrong: restraint makes the child cry and tense, so the exam is useless and distressing. C is wrong: imaging does not replace clinical examination. D is wrong: sedation is not needed and can mask signs. E is wrong: you can palpate between cries and during inspiration; waiting hours delays care.

2. C · A normal-sized liver displaced downward by hyperinflated lungs

C is correct: the span (5 cm) is normal for age; the low upper border shows the liver is pushed down by hyperinflation, a common finding in bronchiolitis and asthma. A is wrong: heart failure gives an enlarged, often tender, rounded liver with an increased span, plus gallop and tachycardia. B is wrong: hepatitis enlarges the liver (increased span) and usually causes jaundice. D is wrong: a tumour causes a firm or hard, enlarged, often irregular liver. E is wrong: glycogen storage disease causes massive hepatomegaly with a large span and hypoglycaemia.

3. B · Spleen

B is correct: a notch, movement toward the RIF with inspiration, inability to get above it, dullness and no ballottement are the classic features of the spleen. A is wrong: a kidney is ballotable, you can get above it, and it is usually resonant (colon in front). C is wrong: faecal masses are indentable, in the left iliac fossa, and do not have a notch. D is wrong: neuroblastoma is a hard, irregular, fixed mass that often crosses the midline. E is wrong: an ovarian cyst arises from the pelvis, and you can get above it but not below.

4. D · Wilms tumour (nephroblastoma)

D is correct: a well child aged 2–5 with a smooth flank mass that is ballotable and does not cross the midline, plus hypertension and haematuria, is classic Wilms tumour. Palpate gently (rupture risk), ultrasound, and refer. A is wrong: neuroblastoma usually makes the child unwell, is irregular, and often crosses the midline. B is wrong: abdominal Burkitt usually presents in older children (4–9 y) with rapid growth and ascites, not haematuria and a ballotable renal mass. C is wrong: hydronephrosis can be ballotable, but hypertension and haematuria in a solid-feeling mass favour Wilms. E is wrong: the spleen is on the left and not ballotable.

5. A · Urgent surgical exploration (with analgesia and nil by mouth) without waiting for imaging

A is correct: these are the classic signs of testicular torsion; viability is 80–100% if detorted within 4–6 hours, so surgical exploration must not be delayed for imaging (RCH). B is wrong: delaying to the morning loses the testis; ultrasound is only for low-probability cases after surgical review. C is wrong: epididymo-orchitis is gradual, with a preserved cremasteric reflex, and antibiotics will not save a twisted testis. D is wrong: that is management of torsion of the appendix testis, with a blue dot and normal reflex. E is wrong: waiting for urine culture delays surgery.

6. E · Reassure: this is a retractile testis; review yearly

E is correct: a testis that can be brought to the bottom of the scrotum and stays there is retractile, a normal finding from about 6 months due to an active cremasteric reflex; review periodically because a few ascend later. A is wrong: orchidopexy is for undescended testes. B is wrong: hormonal treatment is not recommended. D is wrong: the testis has been found; imaging is unnecessary. C is wrong: a unilateral retractile testis with normal genitalia does not suggest a disorder of sex development.

7. D · Refer for herniotomy soon (within weeks) and teach the parents the signs of incarceration

D is correct: inability to get above the swelling and reducibility indicate an inguinal hernia; infants have a high risk of incarceration, so all need prompt elective herniotomy, and parents must know the danger signs (irreducible, tender, vomiting). A is wrong: observation to 1–2 years is for hydroceles (you can get above them and they transilluminate). B is wrong: aspiration has no role and risks bowel injury. C is wrong: trusses are not used in children. E is wrong: leaving a hernia unrepaired risks strangulation.

8. E · Reassure: a non-retractile foreskin is normal at this age; clean the outside only and never force it back

E is correct: a non-retractile foreskin is normal (about 10% retractable at 1 year, 50% by 10 years, 99% by 17); ballooning is usually harmless (topical betamethasone only if problematic). A is wrong: forced retraction causes tears, scarring and true phimosis or paraphimosis. B is wrong: circumcision is not indicated for physiological non-retraction. D is wrong: there is no infection. C is wrong: instrumental dilation is traumatic and unnecessary.

9. D · Do not circumcise; refer to paediatric surgery/urology for hypospadias repair, which uses the foreskin

D is correct: ventral meatus, hooded foreskin and chordee are hypospadias; the foreskin is needed for reconstruction, so circumcision must wait until after repair (usually 6–18 months). A is wrong: circumcision removes tissue needed for repair. B is wrong: any circumcision before surgical assessment is inappropriate. C is wrong: hypospadias needs surgical assessment (spraying stream, chordee, fertility). E is wrong: isolated distal hypospadias with both testes descended does not routinely need karyotyping; it is indicated with undescended testes or ambiguous genitalia.

10. A · Stool microscopy (Kato-Katz) for Schistosoma mansoni eggs and abdominal ultrasound for periportal fibrosis

A is correct: hepatosplenomegaly with portal hypertension in a lakeside child is typical of hepatosplenic S. mansoni; stool eggs and ultrasound periportal ('pipe-stem') fibrosis confirm it; treat with praziquantel 40 mg/kg. B is wrong: SCD causes splenomegaly but not portal hypertension with collateral veins. C is wrong: rK39 tests for visceral leishmaniasis, found mainly in Karamoja/north-east. D is wrong: marrow is for suspected leukaemia or leishmaniasis, not first-line here. E is wrong: S. haematobium causes urinary disease (terminal haematuria), not portal hypertension.

11. C · Endomyocardial fibrosis or constrictive pericarditis

C is correct: ascites out of proportion to peripheral oedema with a very high JVP and soft heart sounds suggests restrictive filling: endomyocardial fibrosis (still seen in Uganda) or constrictive (often TB) pericarditis; echo is needed. A is wrong: nephrotic syndrome has heavy proteinuria and low albumin, with periorbital oedema. B is wrong: kwashiorkor causes peripheral oedema and skin changes, not a raised JVP. D is wrong: Burkitt causes a mass with ascites but not a raised JVP. E is wrong: cirrhosis causes ascites with a normal or low JVP, splenomegaly and stigmata of liver disease.

12. E · Inspect only, in the frog-leg position with the mother present and a chaperone, take a swab from the introitus, and involve a senior because of possible abuse

E is correct: prepubertal girls have external inspection only (frog-leg or on the mother's lap), with consent/assent and a chaperone; a frightened child with discharge raises the possibility of sexual abuse, so a senior and child protection must be involved, with time-sensitive care (PEP within 72 h) if indicated. A is wrong: speculum examination is not done in prepubertal girls on the ward. B is wrong: digital examination is inappropriate and traumatic. C is wrong: restraint is distressing and may retraumatise an abused child. D is wrong: it misses possible abuse, STI or foreign body and fails safeguarding duties.

13. B · A firm, mobile, olive-sized mass in the epigastrium/right upper quadrant

B is correct: hypertrophic pyloric stenosis presents at 2–8 weeks with projectile non-bilious vomiting, visible peristalsis left to right, and a palpable 'olive' best felt during a test feed (ultrasound confirms); check for hypochloraemic hypokalaemic alkalosis. A is wrong: a sausage-shaped mass suggests intussusception (older infants, colicky pain, blood in stool). C is wrong: a ballotable mass is renal. D is wrong: generalised distension with absent bowel sounds suggests ileus or peritonitis. E is wrong: RIF tenderness suggests appendicitis, rare at this age.

14. C · Serum sodium, potassium and blood glucose

C is correct: an apparently male baby with bilateral impalpable gonads and hypospadias may be a virilised 46,XX infant with congenital adrenal hyperplasia; vomiting in week 2 suggests a salt-wasting crisis (low Na, high K, hypoglycaemia), which is life-threatening and needs saline, glucose and hydrocortisone. A is wrong: imaging the gonads is useful later, not first. B is wrong: hormone levels (17-OHP, karyotype) are needed but are not immediately life-saving. D is wrong: there is no sign of obstruction. E is wrong: UTI is a differential for vomiting, but the genital findings point to CAH, which must be excluded urgently.

15. B · Urgent blood transfusion and circulatory support for acute splenic sequestration

B is correct: an acutely enlarging spleen with a sharp Hb fall and shock is acute splenic sequestration, which can kill within hours; restore circulating volume and transfuse urgently (cautiously, e.g. 10 mL/kg packed cells, because blood is released from the spleen as it shrinks), and teach parents to palpate the spleen. A is wrong: this is an emergency. C is wrong: splenectomy is considered after recurrent episodes, not as the first treatment of a shocked child. D is wrong: analgesia alone ignores life-threatening anaemia and shock. E is wrong: test for malaria, but do not delay transfusion.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: OSCE station: examine the abdomen of a 6-year-old with a mass

OSCE station at UCU: 'Opio, a 6-year-old boy, has a 3-week history of abdominal swelling and weight loss. Examine his abdomen and present your findings.' His mother is present.

1. How do you start the station?

> Introduce yourself to Opio and his mother, confirm name and age, explain in simple words ('I want to look at and feel your tummy'), get **consent and assent**, offer a **chaperone** for the genital exam, **wash and warm hands**, position him **supine with one pillow, arms by sides, knees slightly flexed**, kneel to his level, and expose from the nipples to the knees (underpants on until needed).

2. What do you look for in the general examination?

> Well or unwell; **growth** (weight 17 kg, height 112 cm, MUAC); **pallor**, jaundice; **jaw swelling** (Burkitt), periorbital bruising (neuroblastoma); mouth (thrush), **lymph nodes** (cervical, axillary, inguinal), **petechiae/bruises**; oedema; vital signs including **BP**.

3. Describe your inspection, palpation and percussion findings in an organised way.

> Inspection: **asymmetrical distension**, fullness in the right lower quadrant, no dilated veins, umbilicus everted. Palpation (starting away from the mass, watching his face, **gently**): a **14 × 10 cm hard, irregular, non-tender mass** in the RIF/periumbilical region, **crosses the midline**, does not move with respiration, can get above it, liver and spleen not separately palpable, kidneys not ballotable. Percussion: dull over the mass, **shifting dullness present**. Auscultation: normal bowel sounds.

4. How do you complete the examination?

> Groins (no hernia, inguinal nodes), **both testes descended**, normal penis, perianal inspection normal; **BP 104/66** (normal); **urine dipstick** (no blood or protein); **plot growth** (WAZ –1.6); examine chest (dullness/effusion?) and the jaw.

5. Present your findings and differential.

> 'Opio, 6, has a rapidly enlarging, hard, irregular intra-abdominal mass crossing the midline with moderate ascites, weight loss and pallor, with normal BP and urine. The most likely diagnosis is **abdominal Burkitt lymphoma**; differentials are **neuroblastoma**, **abdominal TB** with matted nodes, and Wilms tumour (less likely: crosses the midline, no haematuria, normal BP).'

6. The examiner asks for your next steps.

> **Ultrasound**, FBC, **LDH, uric acid, creatinine, K, phosphate, calcium** (tumour lysis), HIV test, malaria test, chest X-ray; tissue diagnosis (FNA of mass or **ascitic fluid cytology**); start **IV hydration and allopurinol**; urgent referral to the **Uganda Cancer Institute** (or Lacor). Explain to the mother in plain language and support her.

Take-away: Approach the child first, examine gently and systematically, describe a mass with a full checklist, always complete with groins, genitalia, BP, urine and growth, and present a reasoned differential tied to the Ugandan setting.

Case 2: 'Stomach ache' in a 12-year-old boy

At 23:00, Brian, a 12-year-old boy, is brought to the emergency unit of a regional referral hospital with lower abdominal pain and vomiting that started suddenly 2 hours ago while playing football. His mother says he is 'shy' and did not want to be examined.

1. What is your initial approach?

> ETAT: alert, HR 112, BP 118/72, afebrile. Give **analgesia** (paracetamol 15 mg/kg; morphine if severe, per senior). Take a focused history: sudden onset, **vomiting**, no diarrhoea, no urinary symptoms. Explain kindly that you need to examine his tummy **and his private parts**, with his consent, a **chaperone** (male nurse if preferred) and privacy.

2. What do you find?

> Abdomen: mild lower abdominal tenderness, no guarding, normal bowel sounds. **Left hemiscrotum red and swollen**; the **left testis is high-riding with a horizontal lie and exquisitely tender**; the **cremasteric reflex is absent** on the left; the right testis is normal. No hernia.

3. What is the diagnosis and what are the differentials?

> **Left testicular torsion** until proven otherwise. Differentials: torsion of the appendix testis (blue dot, upper-pole tenderness, reflex present), epididymo-orchitis (gradual, fever, dysuria, reflex present), incarcerated hernia, trauma, idiopathic scrotal oedema, HSP.

4. What do you do now?

> **Nil by mouth**, IV access, analgesia, **call the surgeon immediately** for exploration; **do not delay for ultrasound** (time = testis: 80–100% viability within 4–6 h). Consent from the mother for scrotal exploration, detorsion and **bilateral orchidopexy** (orchidectomy if non-viable).

5. In theatre the testis is dusky but pinks up after detorsion. What follow-up and counselling are needed?

> Fixation of **both testes**; review for testicular atrophy; explain to Brian and his mother that **future testicular pain must be seen immediately**; reassure about fertility (one healthy testis is usually enough); teach testicular self-examination as he grows.

6. What lesson about examining boys with abdominal pain?

> **Always examine the scrotum** in a boy with abdominal pain or vomiting: shyness or embarrassment makes adolescents under-report scrotal pain, and torsion often presents as 'stomach ache'.

Take-away: In any boy with abdominal pain, examine the testes; torsion is a surgical emergency (high, horizontal testis, absent cremasteric reflex) and must go to theatre within hours without waiting for imaging.

Case 3: Splenomegaly in a child from northern Uganda

Akello, an 8-year-old girl from Kotido (Karamoja), is admitted to Moroto Regional Referral Hospital with 2 months of intermittent fever, weight loss and a 'big tummy'. She has been treated for malaria three times this year without improvement.

1. What will you look for on general examination?

> **Wasting** (weight, height, MUAC, BMI-for-age), **pallor**, jaundice, **lymphadenopathy**, **bleeding** (epistaxis, gum bleeding, petechiae), skin darkening, oedema, signs of **HIV** (thrush, parotids), TB contacts, temperature and vitals.

2. How will you examine and record the spleen and liver?

> Palpate from the **right iliac fossa** toward the left hypochondrium; spleen edge felt **14 cm below the left costal margin** along its axis, extending past the umbilicus toward the RIF = **Hackett grade 4–5**, firm, non-tender, notch felt. Liver: 3 cm below RCM, span 10 cm. No ascites.

3. What is your differential diagnosis?

> **Visceral leishmaniasis (kala-azar)** (Karamoja, fever, wasting, massive spleen, malaria treatment failure); **hyperreactive malarial splenomegaly**; **chronic malaria**; **schistosomiasis** with portal hypertension; **leukaemia/lymphoma**; **HIV/TB**; brucellosis (pastoralist area).

4. What investigations will you request?

> **rK39 rapid test**; **FBC** (pancytopenia in kala-azar or hypersplenism; blasts in leukaemia); **malaria smear**; **HIV test**; **splenic or bone marrow aspirate** for Leishmania amastigotes if rK39 positive/unclear (by trained staff, with platelets checked); ultrasound; stool Kato-Katz; CXR/Xpert if TB suspected; brucella serology.

5. rK39 is positive and bone marrow shows Leishman–Donovan bodies. What next?

> **Visceral leishmaniasis**: refer/transfer to a **VL treatment centre** (e.g. Amudat Hospital or Kacheliba HC IV) for combination therapy per national VL guidelines (e.g. SSG + paromomycin), nutritional rehabilitation, treat anaemia, test for HIV co-infection, and monitor spleen size and blood counts for response.

6. How do you present this child?

> 'Akello, 8 years, from Kotido, wasted (BMI-for-age Z −2.6), with 2 months of fever unresponsive to antimalarials, pallor, and **massive splenomegaly (14 cm, Hackett 4–5)** with mild hepatomegaly; rK39 positive and LD bodies on marrow: **visceral leishmaniasis** with anaemia and wasting. Plan: VL treatment at a specialist centre, nutrition, transfusion if indicated, HIV test.'

Take-away: Measure and grade a big spleen properly, and let geography guide the differential: in Karamoja, fever and massive splenomegaly that do not respond to antimalarials mean visceral leishmaniasis until proven otherwise.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.